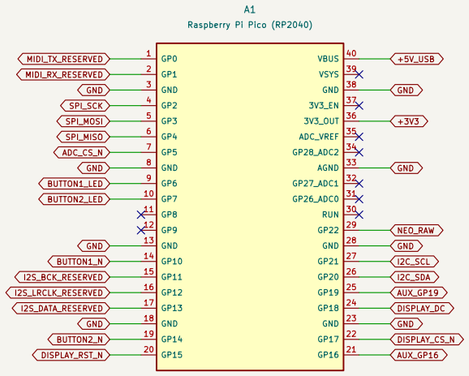
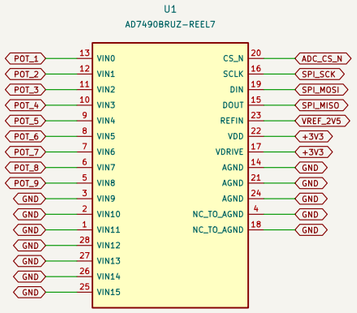
MIDI MACHINES / CONTROL BOARD

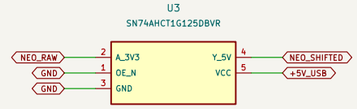
Nine illuminated ALPS controls + two momentary buttons. Display module to be selected.







REF: 2.5 V also excites the pots.



ADC: RANGE=1, straight binary; scan VIN0..VIN8.

Unused analog inputs and ADC NC pins tied to AGND.





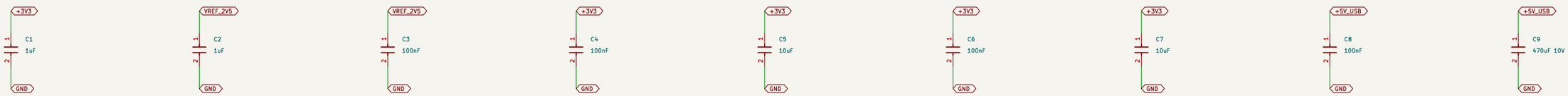




USB on the Pico supplies this draft.

Full-white pixels: ~324 mA + Pico/ADC/display load.

Review USB current declaration; do not inject 5 V while USB is attached.



C1/C2: reference C3: REFIN C4/C5: VDD C6/C7: VDRIVE C8: buffer C9: LED bulk (part TBD)

3 x 3 illuminated controls

File: Controls.kicad_sch

Two buttons + display reservations

File: Buttons_Display.kicad_sch

Prototype schematic; verify mechanical footprints and power budget before PCB layout.

MIDI Machines

Sheet: /

File: MIDI_Machines_Control_Board.kicad_sch

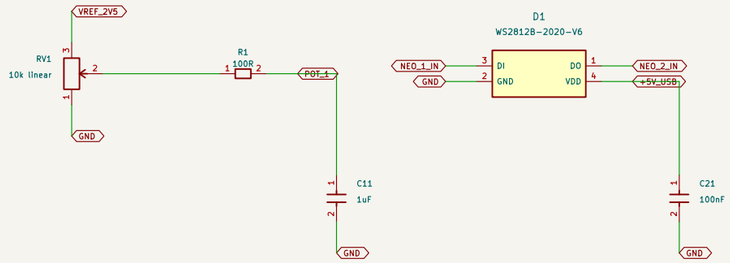
Title: Control board — Pico, ADC and power

Size: A3	Date: 2026-11-13	**Rev: Draft A**
KiCad E.D.A. 10.0.7		Id: 1/3

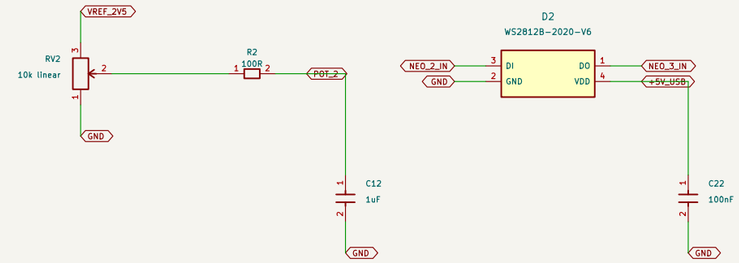
3 x 3 CONTROL GRID / ROW-MAJOR ORDER

Each ALPS shaft receives one separate WS2812B-2020-V6. Physical LED clearance must be verified.

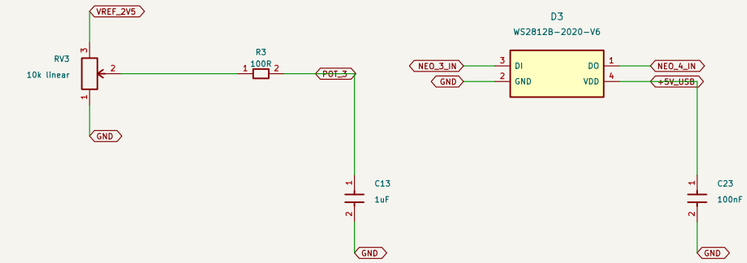
POSITION 1 / ADC VIN0



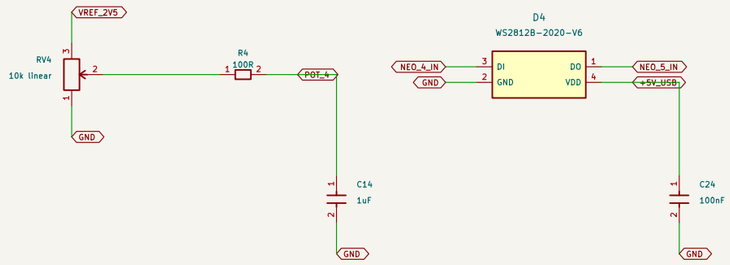
POSITION 2 / ADC VIN1



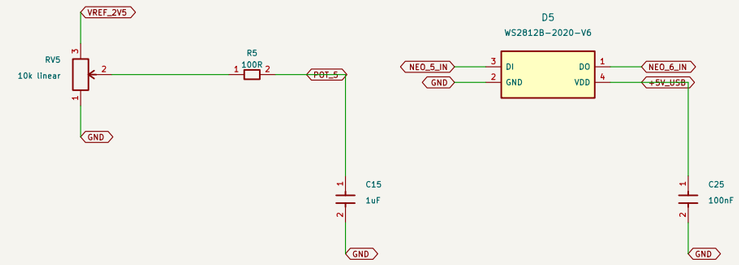
POSITION 3 / ADC VIN2



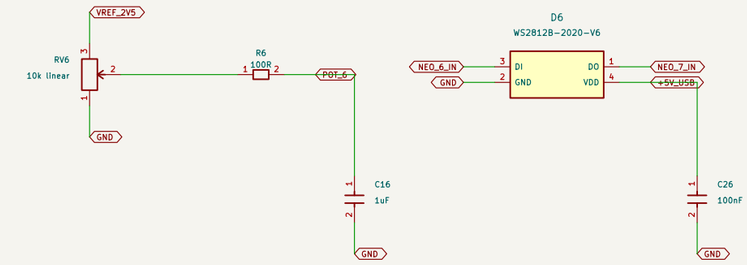
POSITION 4 / ADC VIN3



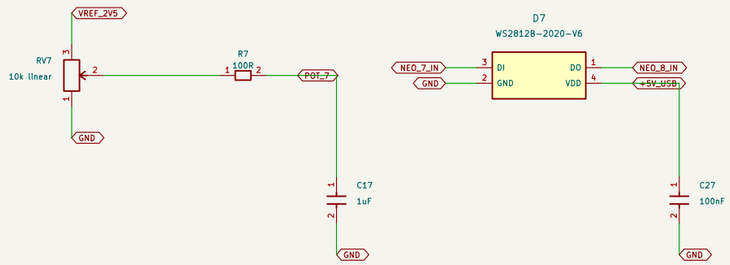
POSITION 5 / ADC VIN4



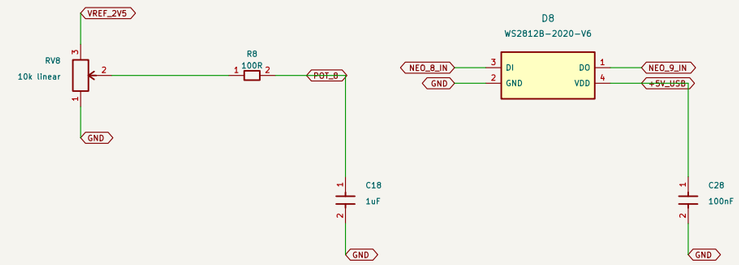
POSITION 6 / ADC VIN5



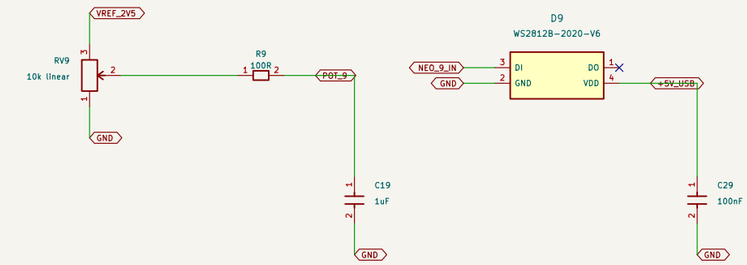
POSITION 7 / ADC VIN6



POSITION 8 / ADC VIN7



POSITION 9 / ADC VIN8



Pixels chain D1 -> D2 -> … -> D9; D9 DO intentionally unused. Data is shifted to 5 V on the root sheet.

Prototype schematic; verify mechanical footprints and power budget before PCB layout.

MIDI Machines

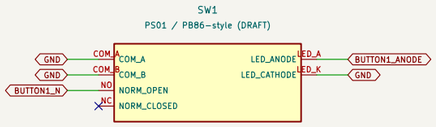
Sheet: /3 x 3 illuminated controls/
File: Controls.kicad_sch

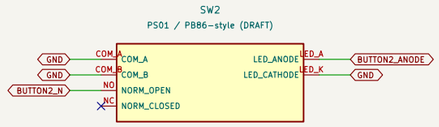
Title: Control board – 3 x 3 illuminated potentiometers

Size: A3	Date: 2026-11-13	**Rev: Draft A**
KiCad E.D.A. 10.0.7		Id: 2/3

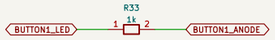
TWO ILLUMINATED BUTTONS / FUTURE DISPLAY

Switches are Tayda PS01 / A-7701 candidates. Logical pin names await physical pad verification.









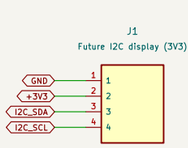


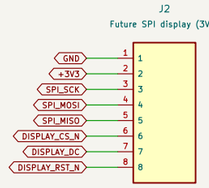


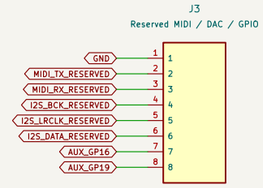
Active-low inputs with 10k pull-ups; firmware debounce required.

Red LEDs: ~1 mA through 1k from Pico GP6/GP7. Not NeoPixels. Verify polarity before assigning a footprint.

DISPLAY CONNECTIONS — MODULE, VOLTAGE AND FOOTPRINT STILL TO BE SELECTED







J1/J2 are provisional breakout pin orders, not pin-compatible guarantees for a display module.

Add I2C pull-ups if the chosen module does not provide them. SPI display uses a separate CS on GP17.

No OLED/LCD controller, backlight driver, display mounting, or 5 V display interface is selected yet.

J3 exposes logic-level signals only; DIN MIDI isolation/drivers and the I2S DAC remain external.

Prototype schematic; verify mechanical footprints and power budget before PCB layout.

MIDI Machines

Sheet: /Two buttons + display reservations/
File: Buttons_Display.kicad_sch

Title: Control board — buttons and future display

Size: A3	Date: 2026-11-13	**Rev: Draft A**
KiCad E.D.A. 10.0.7		Id: 3/3